LESSON PLAN

BASIC ELECTRICAL (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	29	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Fuel system, general design requirements, pressure and gravity system and description of a typical basic fuel system
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Fuel system, general design requirements, pressure and gravity system and description of a typical basic fuel system
5. Specific Objectives	23.1.1.1 Trainee will be able to understand the construction and requirements of typical aircraft fuel system and its components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of fuel system schematic diagrams; Instructor supervision of diagram interpretation exercises.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the aircraft fuel system?
(b) Name two common methods used to deliver fuel to the engine.
(c) Why is fuel system reliability critical for flight safety?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• General design requirements for fuel systems • Safety features in fuel system design • Material selection for fuel tanks and lines • Corrosion prevention and contamination control • Regulatory standards for fuel system construction	L & Ds	PPT, IB	15 min
(b)	• Gravity feed fuel system operation • Pressure feed fuel system operation • Fuel pump types and locations • Fuel strainer and filter function • Fuel quantity indication system basics	L & Ds	PPT, PDF	20 min
(c)	• Description of a typical basic fuel system • Fuel tank construction and arrangement • Fuel lines and fittings specifications • Fuel venting and de-icing systems • Emergency fuel shutoff procedures	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a gravity feed and a pressure feed system.
(b) Identify the key components found in a typical basic fuel system.
(c) Describe the purpose of the fuel strainer in the system.

12. Home Assignment and Follow Up	Prepare a brief note comparing the advantages and disadvantages of gravity and pressure fuel systems, referencing the diagrams in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Types of tanks, constructional details and arrangements in air frame.	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min